SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Annexure A

Debugging & Optimisation Task

Code of Conduct:

I P. Greeshma (192424418) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Greeshma

20/20

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code significantly by improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation

		and documentation			

Scenario 1 :-

Given Network . 192 168 100 .0/24

Subnet Assigned

Department	Subnet	Network Address	Broadcast Address	Valid Host Range	Usable Hosts
School of computing	/25	192.168.100.0	192.168.100.127	192.168.100.1 192.168.100.126	126
School of Electronics	/26	192.168.100.128	192.168.100.191	192.168.100.129 192.168.100.190	62
School of Mechanical Engineering	/27	192.168.100.192	192.168.100.223	192.168.100.193 - 192.168.100.222	30

Remaining unused IP Range

Network block = 192.168.100.0 - 192.168.100.255

used till 192.168.100.223

Remaining unused Range:

192.168.100.224 - 192.168.100.255

Scenario 2

Given Network : 10.10.0.0/16

Department wise Allocation

Department	subnet	Network Address	Broadcast Address	Host Range	Us[illegible] Hosts
Software Development	/24	10.10.0.0	10.10.255.0	10.10.0.1 10.10.0.254	254
Quality Assurance	/25	10.10.1.0	10.10.1.127	10.10.1.1 10.10.1.126	126
Human Resources	/26	10.10.1.128	10.10.1.191.	10.10.1.129 10.10.1.190	62
Executive Management	/27	10.10.1.192	10.10.1.223	10.10.1.193 10.10.1.222	30

Remaining unused IP Range :-

unused addresses begin from

10.10.1.224

up to

10.10.255.255

Scenario 3

Given IPV6 Address.

2001 : 0db8 : abcd : 0000 : 0000 : 0000 : 0000 : 0000 / 64

1. compressed Notation

2001 : db8 : abcd : : /64

2. Expanded Form

2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network prefix (/64)

2001:db8:abcd:0000::/64

4. Number of Host Addresses

Host bits = 128 - 64 = 64

Number of hosts 2^{64} = 18,446,744,073,709,551,616

5. Total possible Host Addresses

18,446,744,073,709,551,616

Scenario 4

Routing table

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

Destination IP

192.168.2.45

Answers:-

1. Destination subnet

192.168.2.0/24

2. Matching Routing Table Entry

192.168.2.0/24

3. Next Hop Interface:-

The packet is forwarded through the interface connected to the

192.168.2.0/24

Network.

4. Default Route:-

If no entry Matches

0.0.0.0/0

(Default Gateway) is used.

Scenario-5

Administration LAN :- 192.168.10.0/26

Finance LAN :- 192.168.10.64/26.

1. Network & Broadcast Addresses

LAN	Network Address	Broadcast Address
Adminstration	192.168.10.0	192.168.10.63
Finance	192.168.10.64	192.168.10.127

2. Valid Host Range:-

Adminstration:- 192.168.10.1 – 192.168.10.62

Finance:- 192.168.10.65 - 192.168.10.126

3. Sugested IP Address:-

Adminstration	Finance
PC1 → 192.168.10.2	Pc1 → 192.168.10.66
PC2 → 192.168.10.3	PC2 → 192.168.10.67
PC3 → 192.168.10.4	PC3 → 192.168.10.68

4. Default Gateway

Adminstration	192.168.10.1
Finance	192.168.10.65